

Hadoop Installation on Ubuntu Operating System

Step-by-Step Practical Guide for Single Node Setup

1. Install Java JDK 8

```
sudo apt update
sudo apt install openjdk-8-jdk
java -version
```

2. Install SSH

```
sudo apt-get install ssh
ssh localhost
```

3. Download Hadoop

```
Download Hadoop 3.2.3 from:
https://hadoop.apache.org/

Extract:
tar -zxvf ~/Downloads/hadoop-3.2.3.tar.gz
```

4. Configure Environment Variables

```
Open ~/.bashrc and add:

export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export HADOOP_HOME=~/hadoop-3.2.3/
export PATH=$PATH:$HADOOP_HOME/bin
export PATH=$PATH:$HADOOP_HOME/sbin
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop
```

5. Configure Hadoop Environment

```
cd hadoop-3.2.3/etc/hadoop
nano hadoop-env.sh

Add:
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
```

6. Configure core-site.xml

```
<configuration>
<property>
<name>fs.defaultFS</name>
<value>hdfs://localhost:9000</value>
</property>
</configuration>
```

7. Configure hdfs-site.xml

```
<configuration>
<property>
<name>dfs.replication</name>
<value>1</value>
</property>
</configuration>
```

8. Configure mapred-site.xml

```
<configuration>
<property>
<name>mapreduce.framework.name</name>
<value>yarn</value>
</property>
</configuration>
```

9. Configure yarn-site.xml

```
<configuration>
<property>
<name>yarn.nodemanager.aux-services</name>
<value>mapreduce_shuffle</value>
</property>
</configuration>
```

10. Configure SSH Key

```
ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
chmod 0600 ~/.ssh/authorized_keys
```

11. Format NameNode

```
hadoop-3.2.3/bin/hdfs namenode -format
```

12. Start Hadoop

```
start-all.sh
jps
```

13. Access Hadoop UI

```
http://localhost:9870
```

14. HDFS Commands

```
hadoop fs -mkdir /user
hadoop fs -mkdir /user/arjun.gautam
touch demo.csv
hadoop fs -put demo.csv /user/arjun.gautam
```

15. Stop Hadoop

```
stop-all.sh
```

Important Viva Questions

Question	Answer
What is Hadoop?	An open-source framework for distributed storage and processing of big data.
What is HDFS?	Hadoop Distributed File System used to store large data.
What is NameNode?	Manages metadata and file system information.
What is DataNode?	Stores actual data blocks.
What is YARN?	Manages cluster resources and job scheduling.

Conclusion: Hadoop can be successfully installed and configured on Ubuntu using single node setup. This setup helps in learning HDFS, MapReduce, and Hadoop ecosystem practically.